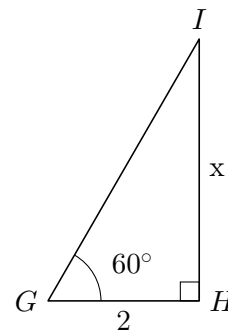
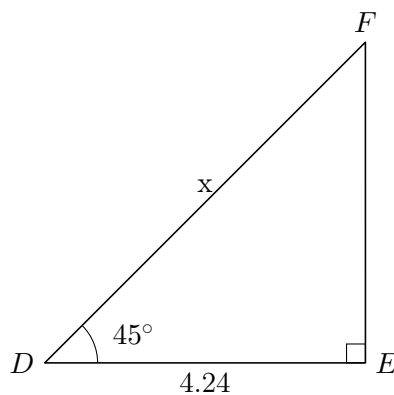
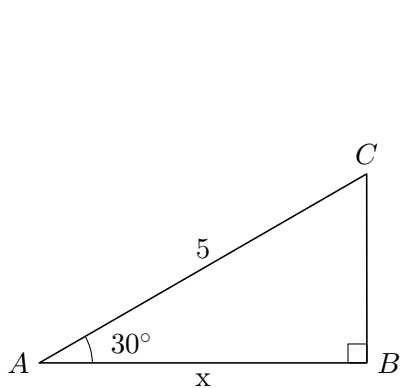


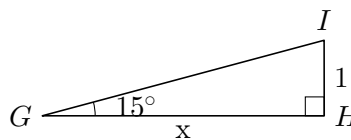
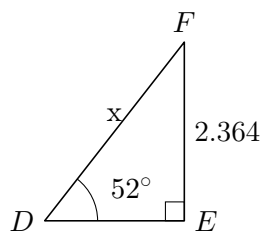
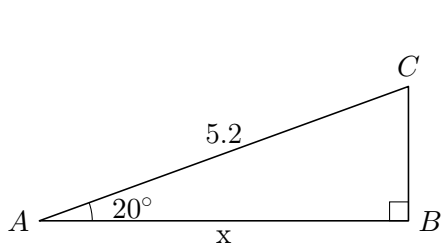
HW 1.4b

Name(s): _____

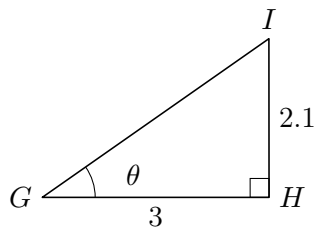
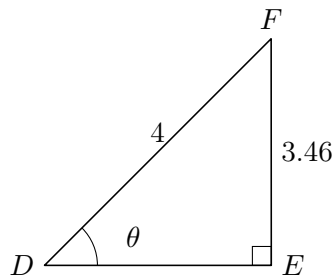
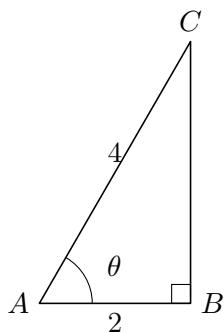
1. Use trig to solve for x .



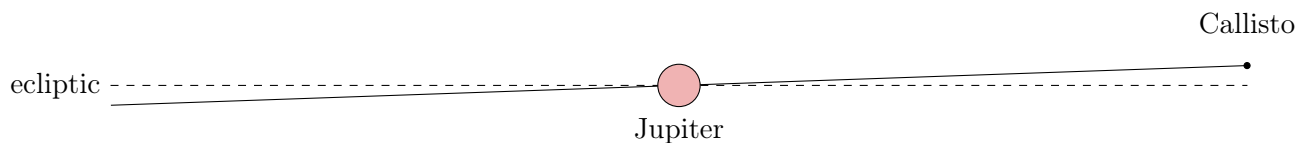
2. Use trig to solve for x



3. Find the unknown angle, θ

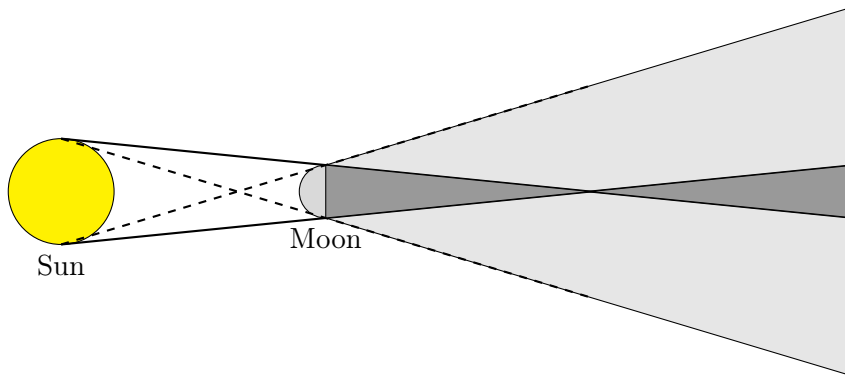


4. Lunar eclipses are more common than solar eclipses because _____.
5. Solar eclipses take place when the Moon is in the _____ phase.
6. During a lunar eclipse, _____
 - A. the Moon is closer to the Sun than it is to the Earth
 - B. the Earth is closer to the Sun than it is to the Moon
 - C. the Earth's shadow passes over the Moon
 - D. the Moon comes directly between the Sun and the Earth
 - E. the Earth has no Moon
7. In order for a total solar eclipse to happen, what three things must happen at the same time?
8. Why are solar eclipses sometimes total, and other times partial or annular?
9. Jupiter has a radius of about 70,000 km. Its moon, Callisto orbits at a distance of 1,880,000 km in a plane inclined by 2 degrees to the ecliptic. The drawing is to scale, except the size of Callisto is 4x its actual in order to be able to see it.



- A. Use trig to figure out the maximum height above the ecliptic that Callisto ever is.
- B. Does Callisto ever fail to have a Solar eclipse when its at New phase?

10. Label the parts of the shadow, and draw in an Earth so there is a total solar eclipse.



11. There is an eclipse season about every _____ months, where we get a Lunar and solar eclipse of some type.
12. Total eclipses happen somewhere on Earth every _____ months. But only happen about once every _____ years for any particular spot on the Earth.
13. Lunar eclipses are actually (which color) _____. This is due to light _____ through the Earth's atmosphere and then landing on the moon.
14. In an annular eclipse the sun is in which part of the Earth's shadow?
15. Label the kind of eclipse shown below. For the total eclipse label the corona of the sun, and Bailey's beads.

